

TeamMatch

A Recommender System for Optimal Employee-to-Project Allocation

Group Project Brief · Recommender Systems (ESADE, Prof. Marc Torrens) · Working draft for the team

The idea in one line: a system that recommends which people should go on which project teams inside a large engineering organisation, then optimises the overall allocation so every project gets a strong, well-balanced team — not just the projects that happen to grab the best people first.

This is a **business-opportunity** project, not a deployment. Our grade rewards the strength of the business case and the impact story (60%) far more than the code (20%), so this brief focuses on *why this matters*, *how it would work*, and *how we'd prove it creates value*. Data will be synthetic; the example company is **Google's internal software-engineering staffing**.

1. The business problem

Large companies constantly reshuffle people across many simultaneous projects. Today this is mostly done manually by managers using spreadsheets, gut feel and who-knows-who. That is expensive and error-prone:

- **~65% of projects** miss their original time, budget or quality targets; **23%** of delays are blamed on poor resource allocation.
- Mis-staffing is costly: IT project failures alone waste an estimated **\$50–150B/year** in the US, and **~\$122M is lost per \$1B** invested due to poor project performance.
- **High-performing teams hit a 63% success rate vs. just 18%** for poorly composed ones — team make-up is decisive.
- In professional services, a **5–10% utilisation swing moves operating margins by 15–25%**, and Gartner finds **60%+ of leaders struggle** with skills-based allocation.

The market is real and growing: workforce-management software is projected to reach **~\$19B by 2030** (~12% CAGR), and internal-talent-marketplace tools (Gloat, Eightfold) are a multi-billion-dollar category. Companies that use talent-marketplace tech are **far more likely** to hit financial targets and adapt to change.

Who benefits

Stakeholder	What they get
Staffing / resource managers	Faster, defensible allocation decisions; less spreadsheet guesswork.
Employees	Projects that fit their skills <i>and</i> interests → engagement & retention.
Project / team leads	Balanced, capable teams with a guaranteed quality floor.
HR / leadership	Higher utilisation and margin; diversity and fairness built in.

2. Our solution: score, then optimise

TeamMatch works in two stages — this is the core of the recommender system:

Stage 1 — Recommendation	Stage 2 — Optimisation
For every (employee, project) pair , compute a compatibility score (e.g. 0–10) blending several signals.	Assign people to projects to maximise the overall match, subject to constraints so no project is left with a weak team.

The score blends interpretable signals (so it is explainable to the business):

- **Skill fit** — the employee's skills vs. what the project needs (content-based matching).
- **Track record** — how this person, or similar people, performed on similar past projects (collaborative-filtering signal).
- **Working-style / personality fit** — from an internal survey (Big Five or DISC), used for *team complementarity*.
- **Preference** — whether the employee actually wants the project (motivation matters — see Project Aristotle below).

The optimisation then respects hard rules: **project size** limits, a **minimum score per project** (a quality floor, so we never sacrifice one team to make another perfect), and a **diversity band** per team (avoid cloning the same profile, without forcing dysfunctional extremes). This is a constrained assignment / multi-objective optimisation problem.

3. An illustrative scenario (Google)

Suppose Google needs to staff three projects. The recommender first scores every engineer for every project (0–10). A naive “pick the highest score” approach would pile the best people onto Project A and starve Project C. TeamMatch instead optimises across all three, respecting each project's size and a minimum-quality floor:

Engineer	Project A (ML platform)	Project B (Maps)	Project C (Ads)	Assigned
Ada	9	6	7	A
Bilal	8	5	8	C
Chen	7	9	4	B
Dara	6	8	7	B

Bilal scores 8 on both A and C, but is sent to C so that Project C still clears the quality floor — a small individual trade-off for a much better *portfolio* of teams. Scale this from 4 engineers / 3 projects to **thousands × hundreds** and the value of automating it becomes obvious. The diversity constraint additionally prevents, say, four near-identical profiles landing on the same team.

4. Why ours is different (the wedge)

Existing tools (Napta, Mosaic, Deltek, Monday.com) already match people to work on **skills + availability**. Our differentiator is everything they ignore:

- **Team chemistry, not just individual skills.** Google's own **Project Aristotle** (180+ teams studied) found the “*who*” (skills, seniority, personality of individuals) mattered less than *how the team works together* — with **psychological safety** as the #1 driver. A naive skills-matcher misses this; we model team-level fit.
- **Diversity as an explicit objective.** McKinsey links top-quartile diversity to **~39% higher likelihood of financial outperformance**; we build it into the optimisation rather than bolting it on.
- **A quality floor for every project**, instead of blindly maximising the average.
- **A learning loop** that improves the recommender from real outcomes (most tools are one-shot).

Capability	Existing tools	TeamMatch
Skill & availability matching	Yes	Yes
Team-level chemistry / psychological safety	Rare	Core
Diversity as an optimisation objective	Afterthought	Built-in
Minimum-quality floor per project	No	Yes
Learns from realised team outcomes	Limited	Yes

5. How it learns & how we prove value

Feedback loop: after each project we collect labels — supervisor team ratings, peer/360 feedback, client satisfaction (for client-facing work), and objective delivery metrics (on-time, on-budget, defects, velocity). These become the ground truth that recalibrates the scoring weights over time.

KPIs vs. today's manual allocation (anchors the 30% “evaluation & business impact” grade):

- **Business:** utilisation rate, project margin, time-to-staff a project, retention/attrition, rework/overrun rate.
- **Recommendation quality:** score uplift vs. baseline, % of projects meeting the quality floor, team diversity index, ranking metrics (Precision@k / NDCG once feedback exists).
- **Validation:** a backtest — compare TeamMatch's proposed allocations against historical manual ones on realised outcomes (achievable on synthetic data with a hidden “true effectiveness” function).

6. Risks & ethics (a graded section)

Allocation decisions shape careers, so we must address: **bias** in who gets high-visibility projects; the validity and legality of **personality testing**; **privacy** of performance data; and **transparency & employee agency** (give people explanations and some control — the professor's recommended remedy). There is also a risk of **homogenising** teams or locking people into a profile.

7. To decide together next meeting

- Commit to **Google as the worked example** vs. a generic “large engineering org” (Google as illustration).
- Optimisation objective: **average score + min-floor + diversity band** (recommended) vs. maximise the *minimum* team score.
- Personality model: **Big Five** (more scientifically defensible) vs. **DISC** (more corporate-familiar).
- Do we build the **optional synthetic-data prototype** (the “extra mile”), or keep it concept-only?

Sources

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